COMP 3920 Database Systems 1

Chat Messaging System – Unread Messages

Introduction:
Now that you've designed your new chat messaging system, your users have asked for a new feature. They want to know which messages they haven't read yet. In order to give them this feature, you are going to need to modify the database.

Step 1: Change your chat messaging system database to handle storing, for each user and each room, which message(s) are unread.

Change your database as needed to accommodate storing this new information.

HINT: You can assume that if a message is unread, all newer messages are also unread.
HINT2: You shouldn't need to make a big change to the database, a small change to an existing table will likely suffice.

Submit a screenshot of the new ERD. Use the MySQL Workbench Reverse engineer function to produce an ERD with your new design.

Submit your file:
01_ERD_with_Unread_Messages.jpg

Submit

Step 2: Update the data so that everyone has no unread messages.

Hint 1: there is an easy way and a harder way.

Easy Way: The easy way is to set everyone's last read message id to the max of all the message ids from the message table.

Harder Way: Set everyone in each room to the maximum message id for that room.

Step 3: Write a query to list everyone and all the groups they have unread messages in.

Currently, no one should unread messages since we just set everyone to have read all messages.

Step 4: Add the following 4 new messages.

```
INSERT INTO `message`
VALUES
(26,1,'2023-02-06 15:12:45','Perfect! See you all there!'),
(27,15,'2023-02-06 17:05:13','My cat sheds so much!'),
(28,22,'2023-02-06 18:41:23','I want to go to Reno!'),
(29,3,'2023-02-06 20:55:01','I can\'t make it. Have fun!');
```

Step 5: Re-run your unread messages query from to see who has unread messages.

Submit a screenshot for the people (in rooms) with unread messages.

Submit your file:
02_Users_with_Unread_Messages.jpg

Submit

```
#Which user has unread messages in their group?
SELECT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	room_id
1	1
4	1
5	1
6	1
2	4
3	4
6	4
2	6

Result 334

Step 6 (Challenge): List the <u>number of unread messages</u> everyone has in all the groups they have unread messages in.

Submit a screenshot for the with user, the room they are in and the **number of unread messages** in each room.

Submit your file:
03_Number_of _Unread_Messages.jpg

Submit

```
#Count the number of unread messages per user.
SELECT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	room_id	num_messages_behind
1	1	2
4	1	2
5	1	2
6	1	2
2	4	1
3	4	1
6	4	1